Day 075 | Unit 05 | Teacher plan

Represent and Solve Equations and Inequalities

Unit 5 Test

Standards: 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4a, 6.NS.B.4b, 6.EE.B.5, 6.EE.B.7, 6.EE.B.6, 6.EE.B.8, 6.EE.C.9

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

Assessment routine

5 minutes: explain response types without solving any item. 35 minutes: independent assessment. 10 minutes: collect work and note misconceptions for the next lesson. Keep the answer key and worked-example help unavailable during assessment.

Review allocation

Questions 1–4 are previous-unit review (20%). Current-unit questions increase in demand after the review. Unit 1 uses Grade 5 readiness skills because no earlier course unit exists.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 3, 5, 11, 14, 17, 18, 20, 25

Day 075 | Unit 05 | Unit Test

Represent and Solve Equations and Inequalities

Unit 5 Test

Standards: 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4a, 6.NS.B.4b, 6.EE.B.5, 6.EE.B.7, 6.EE.B.6, 6.EE.B.8, 6.EE.C.9

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

Directions

The first 4 questions (20%) review Unit 4. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Write 3^2 as repeated multiplication and evaluate.

02 / Previous unit review | Constructed response

Evaluate 8 + 5 × 2. Show the intermediate steps.

03 / Previous unit review | Constructed response

Find the GCF of 10 and 15. Show factor pairs or prime factors.

04 / Previous unit review | Constructed response

Find the LCM of 2 and 6 by listing multiples.

05 / Constructed response

Does x = 7 solve 3x + 7 = 25? Show substitution.

06 / Constructed response

Solve x + 3 = 5. Check in the original equation.

07 / Constructed response

Solve x - 7 = 5/2. Check in the original equation.

08 / Constructed response

Solve 7 × x = 21/2. Check in the original equation.

09 / Constructed response

Solve x + 4 = 15/2. Check in the original equation.

10 / Constructed response

Solve x ÷ 8 = 7/16. Check in the original equation.

11 / Constructed response

7 identical lengths of rope total 28 m. Let x be the length of one piece. Write and solve an equation, then check.

12 / Multiple select

A club needs at least 12 members. Select all true statements for its membership count n.
A. 11 is an allowed membership count.
B. n > 12
C. 12 is an allowed membership count.
D. n ≥ 12

13 / Multiple select

A club needs at least 15 members. Select all true statements for its membership count n.
A. n > 15
B. 14 is an allowed membership count.
C. n ≥ 15
D. 15 is an allowed membership count.

14 / Multiple select

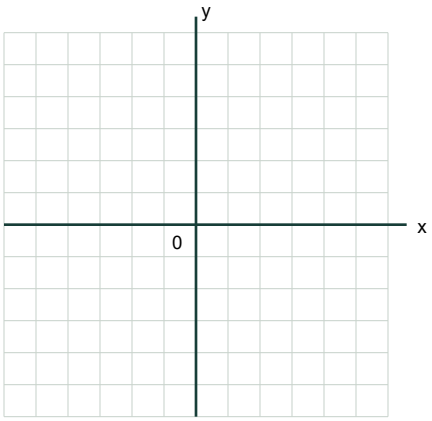
A club needs at least 42 members. Select all true statements for its membership count n.
A. n ≥ 42
B. 41 is an allowed membership count.
C. 42 is an allowed membership count.
D. n > 42

15 / Multiple select

A club needs at least 25 members. Select all true statements for its membership count n.
A. n > 25
B. 24 is an allowed membership count.
C. 25 is an allowed membership count.
D. n ≥ 25

16 / Table completion, equation, and graph

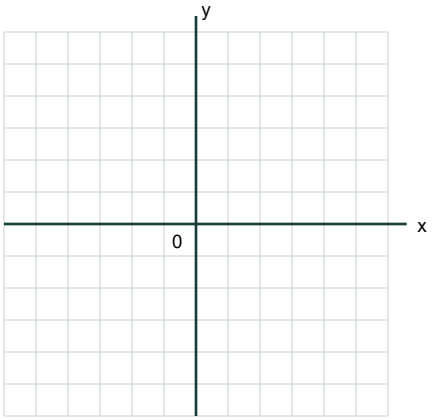
A machine uses 4 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (4, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

17 / Table completion, equation, and graph

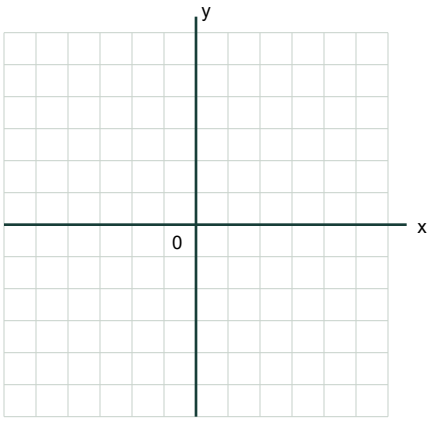
A machine uses 3 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (6, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

18 / Table completion, equation, and graph

A machine uses 6 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (4, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

19 / Multiple select

Select all values of x from the list that make x + 7 > 12 true. Explain why each rejected value fails.

A. 5
B. 4
C. 6
D. 7

20 / Multiple select

Solve x + 5/2 = 7. Select every representation of the solution.

A. 11/2
B. 18/4
C. 13/2
D. 9/2

Day 075 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Unit 5 Test

Standards: 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4a, 6.NS.B.4b, 6.EE.B.5, 6.EE.B.7, 6.EE.B.6, 6.EE.B.8, 6.EE.C.9

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 3 × 3 = 9

The exponent counts 2 factors of 3; multiply all factors.

6.EE.A.1 | Constructed response

2. 18

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 18.

6.EE.A.1, 6.EE.A.2c | Constructed response

3. 5

Common factors: 1, 5; greatest is 5.

6.NS.B.4a | Constructed response

4. 6

The first positive multiple shared by 2 and 6 is 6.

6.NS.B.4b | Constructed response

5. No; left side = 28

3 × 7 + 7 = 28; compare with 25.

6.EE.B.5 | Constructed response

6. x = 2

Subtract 3 on both sides. Then x = 2; substitution gives 5 on each side.

6.EE.B.7 | Constructed response

7. x = 19/2

Add 7 on both sides. Then x = 19/2; substitution gives 5/2 on each side.

6.EE.B.7 | Constructed response

8. x = 3/2

Divide by 7 on both sides. Then x = 3/2; substitution gives 21/2 on each side.

6.EE.B.7 | Constructed response

9. x = 7/2

Subtract 4 on both sides. Then x = 7/2; substitution gives 15/2 on each side.

6.EE.B.7 | Constructed response

10. x = 7/2

Multiply by 8 on both sides. Then x = 7/2; substitution gives 7/16 on each side.

6.EE.B.7 | Constructed response

11. 7x = 28; x = 4 m

Divide the total by 7: 28 ÷ 7 = 4.

6.EE.B.6, 6.EE.B.7 | Constructed response

12. C, D - 12 is an allowed membership count.; n ≥ 12

At least includes the boundary, so n ≥ 12. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

13. C, D - n ≥ 15; 15 is an allowed membership count.

At least includes the boundary, so n ≥ 15. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

14. A, C - n ≥ 42; 42 is an allowed membership count.

At least includes the boundary, so n ≥ 42. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

15. C, D - 25 is an allowed membership count.; n ≥ 25

At least includes the boundary, so n ≥ 25. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

16. (0,0), (1,4), (4,16); independent h; L = 4h

Hours are the input. Multiply each hour value by 4 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

17. (0,0), (1,3), (6,18); independent h; L = 3h

Hours are the input. Multiply each hour value by 3 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

18. (0,0), (1,6), (4,24); independent h; L = 6h

Hours are the input. Multiply each hour value by 6 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

19. C, D - 6; 7

Subtract 7 conceptually: x must exceed 5. Equality at 5 fails because the inequality is strict.

6.EE.B.5 | Multiple select

20. B, D - 18/4; 9/2

Use the inverse operation on both sides. The solution is 9/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select